



Post-Quantum Mathematics

Part 4 - Primitives & Cryptanalysis: Symmetric & Hash-Based Research

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The usual summary of quantum computing's effect on cryptography is that public-key algorithms break and symmetric ones merely need longer keys. That summary is right about AES and wrong about almost everything built on top of it.

Grover's algorithm is the reason for the optimistic half. It searches an unstructured space quadratically faster than any classical method, which halves the effective strength of a symmetric key. Doubling the key restores the margin, and that is the whole of the standard advice.

The complication is Simon's algorithm, which gives an exponential rather than quadratic speedup against constructions with a particular hidden structure. Many standard modes of operation and message authentication codes turn out to have it. Whether that matters depends on a threat model this part has to name up front. The Q2 model assumes an attacker can query the cipher in quantum superposition, which requires access no realistic attacker has today. The results in Section 2 are real mathematics about a setting that may never arrive.

Section 1 covers Grover and the AES question. Section 2 covers the Simon's-algorithm results. Section 3 covers which constructions now have quantum-era security proofs. Section 4 covers hash-based signatures beyond the ratified standard. Section 5 covers the hash functions the ratified standards depend on internally.

1. Grover and the AES-256 Question

This section is short because the answer is settled. The interesting part is that the settled answer comes with a specific number attached.

Current State

Grover's algorithm provides a quadratic speedup against symmetric-key search, halving effective key strength [1]. For AES-256, the classical search space of 2^{256} becomes roughly 2^{128} against a quantum attacker.

A cost of 2^{128} operations remains considered completely infeasible [2]. The NSA's current guidance requires 256-bit AES keys for all classification levels, and treats AES-256 as unchanged from its previous guidance [2]. It characterizes the resulting 128 bits of security against Grover as sufficient for the foreseeable quantum future.



2. Simon's Algorithm and the Q2 Model

This is where the reassuring summary stops holding. Against certain constructions the speedup is exponential rather than quadratic, and the constructions affected are ones in wide use.

Recent Developments

An April 2026 paper demonstrates the first practical implementation of Simon's algorithm for key recovery [13]. It ran on small Even-Mansour ciphers of three and four bits, on real quantum hardware.

The scale is tiny and the significance is not. Until now these attacks existed only on paper. Running one on hardware, even at toy size, moves the work from theory to demonstration.

Current State

Simon's algorithm, applied in the quantum-query model where an attacker can query the cipher in superposition, recovers keys from certain symmetric constructions in a linear number of queries [3]. That is a genuine exponential speedup on those specific constructions, and a different thing entirely from Grover's uniform quadratic one.

A body of work has shown that many symmetric schemes and block-cipher constructions provide substantially less security than expected in this model [4]. They remain fully secure classically.

A 2026 result extends the technique further. It combines an offline variant of Simon's algorithm with a dedicated attack [5]. This breaks block-cipher constructions built from two parallel permutation-based functions, plus one message authentication construction.

3. Quantum Security Proofs for Block Ciphers and MACs

A classical security proof does not automatically survive a quantum attacker. This section covers which constructions have been re-proved and which have been shown to fail.

Recent Developments

A 2026 paper provides the first post-quantum security proofs for a key-length extension scheme, two tweakable block cipher constructions, and most block cipher encryption and authentication modes [14].

Current State

The failures came first. Block-cipher-based message authentication codes including CBC-MAC, GMAC, PMAC and GCM are provably secure classically up to $2^{n/2}$ queries [6]. All have been shown vulnerable to quantum forgery attacks using superposition queries. Their classical security proofs do not carry over.

There are positive results too. A 2026 paper provides post-quantum justification for practical block-cipher-based correlation-resistant hashing, by reducing its security to the multi-key security of a tweakable block cipher [7].

A 2026 survey catalogues which symmetric and block-cipher constructions have quantum-era security reductions and which remain only classically justified [6]. It is a map of the field rather than a new result, and the existence of such a map is itself a useful signal about how uneven the coverage is.

4. Hash-Based Signatures Beyond SLH-DSA

Hash-based signatures rest on nothing but the security of a hash function, which makes them the most conservative signatures available. NIST ratified one stateless scheme and approves two stateful ones, and research continues on all three.

Recent Developments

A February 2026 paper presents a unified configurable hardware architecture supporting key generation and signing for both the stateful schemes and the stateless one [16]. It achieves up to 4.12 times and 10.92 times lower area-time product for stateful signing compared with prior single-scheme designs.

A May 2025 paper introduces new encodings applicable to these schemes, achieving a 20% to 40% improvement in verification cost at the same security level and signature size [15]. It works by mapping non-uniformly into larger hypercubes with controlled collisions.

Current State

NIST specifies two stateful hash-based signature algorithms in SP 800-208: LMS and XMSS, along with their multi-tree variants [8]. Both remain the approved options for software and firmware signing under current guidance as of January 2026, with validation available.

Stateful means the signer must remember which keys have been used, and reusing one destroys security. That burden is why these schemes are confined to signing tasks where a single controlled signer can be guaranteed.

Both schemes offer multiple parameter sets, trading off signature size, public-key size, the total number of signatures a keypair can produce in its lifetime, and verification cost [9]. LMS is generally reported as faster for both signing and verification. A February 2025 benchmark study published reference performance data across the parameter options to support that choice [10].

5. Hash Function Margins

The ratified standards from Part 1 use SHA-3 internally. That makes SHA-3's own quantum margin load-bearing for algorithms most people think of as lattice-based.

Current State

The extendable-output functions SHAKE128 and SHAKE256 are used inside both FIPS 203 and FIPS 204 [12]. Their quantum security margin is therefore directly load-bearing for two ratified standards, not merely a general-purpose hash concern.

One recent result needs careful reading. A December 2025 paper applies Grover's algorithm to a reduced-round variant of Keccak-256, using three rounds rather than the full twenty-four [11]. It reduces preimage-search complexity from a classical $2^{57.8}$ to a theoretical quantum $2^{28.9}$.

The paper's own resource estimate is the reason it is reassuring rather than alarming. It requires 3.2 million physical qubits and 7.47×10^{13} two-qubit gates, with runtime from 43 days to over 2,300 years depending on hardware assumptions [11]. The authors conclude that full SHA-3 remains secure against foreseeable quantum attack, and the result does not apply to the full-round function actually deployed.

Post-quantum algorithms using SHA-256 or SHAKE internally set their parameters with an internal margin [12]. The margin exists specifically so Grover's speedup does not erode security below target levels.

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